A primary + B companion: conditional analytical checks

7 October 2026 — Stage 3 open; S3-G2-Q-T transient-comparison decision

Status and interpretation. These are reviewer deductions and a proposed local mean construction, not traffic simulation or a theorem for the joined multilane reference. Source basis: the point-renewal construction in S12A005, Section 2; the native boundary in RM0010, Section 6 equation (15); and the supplied 2006 motion rule. No new task law is adopted by this document.

1. A: a bounded point-renewal candidate

Let $u, w > 0$ be free speed and backward wave-speed magnitude (m/s), $\kappa > 0$ jam density (vehicle-equivalents/m), $q \geq 0$ an actual insertion rate, and $m_* = 1$ vehicle-equivalent the source descriptor-frequency unit. $h = m_*/q$ is a renewal interval, not a rounding of transferred mass. For fixed own-model entry-speed/kinetic inputs, let $V(s)$ be a nondecreasing free-motion speed, $0 \leq V(s) \leq u$, approaching u, and $X(s) = \int_0^s V(a)\, da$.

In a repeating point-insertion subcase, the next backward characteristic and the current free-motion path intersect at a unique $T = T(h)$:

$$X(T) + wT = wh, \qquad h = m_*/q, \qquad Q = \frac{uw\kappa}{u+w}. \tag{1}$$

The left side is continuous and strictly increasing from zero, so $0 < T \leq h$. Define the candidate renewal mean and its zero-rate extension:

$$C_A(q) = w\kappa\left(1 - \frac{T(h)}{h}\right) \quad (q > 0), \qquad C_A(0) = Q. \tag{2}$$

This extends the local construction to a declared free-motion curve. It does not claim that the curve remains free under all reference interactions. The space-time renewal assumption, fixed entry inputs and one insertion location are explicit approximation conditions.

Since $0 \leq X(T) \leq uT$,

$$\frac{w}{w+u} \leq \frac{T}{h} \leq 1, \qquad 0 \leq C_A(q) \leq Q, \qquad \lim_{q \downarrow 0} C_A(q) = Q. \tag{3}$$

For the limit, $T \to \infty$ and $X(T)/T \to u$ in (1). If $V \equiv u$, $C_A = Q$ at every rate. These are properties of the defined local closure; they are not exact equivalence through transient recovery of old constraints.

Excluded from this proof: distributed insertion positions, downstream restriction of the free path, neighbor interference, heterogeneity and a multilane spatial port. Applying a penalty repeatedly at successive cells is not justified. The restricted point task has not been selected.

2. Conditional sensitivity and unique priority coupling

At smooth points, differentiate (1) at fixed kinetic/entry-speed inputs:

$$T_h = \frac{w}{w + V(T)}, \qquad \frac{dC_A}{dq} = \frac{w\kappa}{m_*}\big(hT_h - T\big) \le 0. \tag{4}$$

The inequality follows because $X(T) \le TV(T)$ for nondecreasing V:
$hT_h = (X(T) + wT)/(w + V(T)) \le T$. Piecewise motion has the same monotonicity; ordinary derivatives are only asserted away from its corners. For a speed parameter b with $X_b(T) \ge 0$,

$$T_b = -\frac{X_b(T)}{w + V(T)}, \qquad \partial_b C_A = \frac{w\kappa X_b(T)}{h(w + V(T))} \ge 0. \tag{5}$$

Along metering changes, entry speed may also change. The total response has both contributions; neither partial sign proves an optimal control policy.

Let $D_r, D_m \ge 0$ be own-state ramp/mainline demands, $0 < p < 1$ fixed ramp priority, and $0 \le R_0 \le Q$ ordinary receiving. All are rates. Define

$$R(q) = \min\{R_0, C_A(q)\}, \qquad G(R) = \min\{D_r, \max\{R - D_m, pR\}\}, \qquad q = G(R(q)). \tag{6}$$

Here birth speed is prescribed from the current own-state support, not obtained by inverting the accepted flow. $R(q)$ is continuous and nonincreasing; G is continuous and nondecreasing. Thus

$$\phi(q) = G(R(q)) - q \text{ is strictly decreasing}, \qquad \phi(0) \ge 0, \quad \phi(D_r) \le 0. \tag{7}$$

There is exactly one root in $[0, D_r]$ (including the singleton $D_r = 0$). Using the same receiving award, mainline service is

$$f_m = \min\{D_m, \max\{R(q) - D_r, (1-p)R(q)\}\}, \qquad q + f_m = \min\{D_r + D_m, R(q)\} \le R_0. \tag{8}$$

Actual accepted flows are debited/credited once in the conserved spatial state. This closes one scalar algebraic loop under its assumptions. It does not prove a multidimensional fixed point when speed, lateral transfers and several capacities are implicit unknowns. No root was numerically solved.

Interpretation. The upstream permission acts at the entrance of a finite ramp approach. Its effect on D_r, the speed inputs and this node occurs after propagation. Replacing q by contemporaneous meter permission in (2) would change the selected actuator.

3. Recovery duration does not determine recovery distance

This is a continuum free-motion illustration of the supplied level-road kinetics, not a replacement of its sampled motion/birth convention. Suppose $v_{\max} > u > v_0 \geq 0$, $a_0 > 0$, and $\lambda = a_0/v_{\max}$:

$$\dot{v} = a_0(1 - v/v_{\max}), \qquad v(t) = v_{\max} - (v_{\max} - v_0)e^{-\lambda t}, \qquad T_{\text{rec}} = \frac{1}{\lambda} \log \frac{v_{\max} - v_0}{v_{\max} - u}. \tag{9}$$

Set $V(t) = v(t)$ until T_{rec} and $V(t) = u$ afterwards. Its free-speed distance deficit d has units of metres:

$$d_{\text{curve}} = \int_0^{T_{\text{rec}}} (u - v(t))\, dt = (u - v_{\max})T_{\text{rec}} + \frac{u - v_0}{\lambda}. \tag{10}$$

A constant acceleration matching only the duration is $a_{\text{eff}} = (u - v_0)/T_{\text{rec}}$. Its linear speed curve gives

$$d_{\text{linear}} = \tfrac{1}{2}(u - v_0)T_{\text{rec}}, \qquad 0 < d_{\text{curve}} < d_{\text{linear}}. \tag{11}$$

Indeed the nonlinear speed curve is strictly concave and lies above its endpoint chord. For small enough insertion rate that the intersection occurs after recovery, $X(T) = uT - d$, and (1)–(2) yield

$$C_A(q) = Q - \frac{w\kappa}{w + u} \frac{q}{m_*}\, d. \tag{12}$$

Consequently equal recovery times need not produce equal renewal service. Matching a_{eff} solely by duration is an additional constitutive approximation. This is not evidence that the nonlinear reference is empirically superior, or that detailed identity is necessary.

Spatial qualification remains separate. The original multilane mean model uses a particular spatial partition and multiple coupled flows. The present point construction supplies no theorem for those interactions. A qualified spatial construction is required before this local result is promoted to the primary task's model. Restricting geometry alone does not resolve the transient-admission issue on page 5.

4. B: trigger plateau and inactive checks

Let D_r, D_m, R_0, Q, p have the meanings above, and $0 < \delta < 1$ be a fixed calibrated drop fraction. The native phenomenological boundary uses

$$R_B = \min\{R_0, Q(1 - \delta\, \mathbf{1}_{\{D_r + D_m > R_0\}})\}, \qquad F_B = f_r + f_m = \min\{D_r + D_m, R_B\}. \tag{13}$$

The same fixed-priority allocation gives $0 \le f_j \le D_j$ and $F_B \le R_0$. There is no new recovery timer. If the trigger stays active, $R_0 \ge Q(1 - \delta)$ and enough demand is present, then

$$F_B = Q(1 - \delta). \tag{14}$$

Changing ramp demand can redistribute this total without changing it. This is a local fixed-regime result; future receiving, availability, trigger timing and exits can still respond. It is not a full-TST invariance theorem.

New conditional onset check. Take a descriptor-free sampled reference and enough target space for the first ordinary update. At that update let

$$R_0 = Q, \quad D_m = Q, \quad D_r = \varepsilon Q, \quad 0 < \varepsilon < p(1 - \delta), \quad \Delta t\, \varepsilon Q < m_*(1 - z), \quad 0 \le z < 1. \tag{15}$$

z is public initial transfer-threshold progress. Both nodes award the full small ramp request. With no other descriptor-generating transfers, the source threshold is not reached during this bin, so no new moving restriction exists to reduce the ordinary reference node. Under these explicit conditions,

$$F_{\text{ordinary}} = Q, \quad F_B = (1 - \delta)Q, \quad f_r^{\text{ordinary}} = f_r^B = \varepsilon Q. \tag{16}$$

This is a local activation/binning mismatch, not a calibrated freeway witness or an end-to-end reachability proof. It does not justify claiming necessary hidden information. The earlier check $D_r = 0, D_m = Q, R_0 = 0.9Q, \delta = 0.2$ also remains $F_B = 0.8Q$ versus ordinary $0.9Q$.

For any model with common arrivals $A(t)$, initial population P_0 and exits $E(t)$, conservation of all waiting/ramp/road mass gives

$$M(t) = P_0 + A(t) - E(t), \qquad J = \int_0^\infty M(t)\, dt, \qquad J_a - J_b = \int_0^\infty \big(E_b(t) - E_a(t)\big)\, dt. \tag{17}$$

Assume finite objectives or a certified tail. Identical complete exits imply identical TST; identical instantaneous merge output alone does not. Joined event integration, clearance, intervention leverage and useful gradients remain unproved. A/B coordination is approved; neither Q-T candidate is selected.

5. A's steady limit does not certify sampled startup

Consider the proposed present-entry point mean, not every possible A closure. Keep $R_0 = Q, D_m = Q$, prescribe a slow own-state birth speed $v_0 < u$, and assume $D_r \geq pQ$. The fixed point in (6) reduces to $q = pC_A(q)$. Its unique root is positive, and $V(s) < u$ for an initial nonzero interval implies

$$q = pC_A(q) > 0, \qquad X(T) < uT, \qquad F_A = C_A(q) < Q. \tag{18}$$

This state can correspond to a slow queued ramp donor; it does not require a small sending demand to imply a slow free-flow speed. Start with no existing descriptors and public threshold progress $z < 1$, no other descriptor-generating transfers, and a sufficiently small admissible bin:

$$\Delta t\, pQ < m_*(1 - z), \qquad F_{\text{ordinary}} = Q > F_A, \qquad F_B = (1 - \delta)Q. \tag{19}$$

No reference descriptor is created in this bin. The ordinary node awards pQ to the ramp; A and B award less, and all remain below the threshold. The reference is unconstrained by descriptor kinetics during this local update, whereas both present-condition service approximations can already penalize it. Target storage and sending feasibility are assumed as in the previous check.

Conclusion and limits. This is a conditional local onset/averaging check, not an end-to-end reachable scenario or refutation of mean models. It shows why $C_A(0) = Q$ and free-speed equality are weaker than the entire sampled reference's transient inactive requirement. Future traffic, old descriptors, spatial interactions and overall TST have not been solved.

Adding an independently evolved event/recovery summary would change A's retained-state contract; accepting the known discrepancy for a definitive predictive rival would change comparator admission. Q-T presents those two boundaries. Neither is adopted, and the reference birth sampling is unchanged.